

Section 17 Solution

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2 Exercises

2.1 Dominant Strategies

Applying the definition: Consider the game represented by

		Player 2		
		L	C	R
Player 1	T	1,2	2,1	3,4
	M	2,1	5,0	2,6
	B	3,7	8,2	5,5

1. Prove or disprove that $s_2 = R$ is a dominant strategy for player 2.

Answer:

$s_2 = R$ is not a dominant strategy for player 2: if Player 1 plays B, then Player 2 strictly prefers L to R. That is, $U_2(B, L) = 7 > 5 = U_2(B, R)$.

2. Prove or disprove that $s_1 = B$ is a dominant strategy for player 1.

Answer:

$s_1 = B$ is a dominant strategy for player 1. We need to check all $s_2 \in \{L, C, R\}$:

$$U_1(B, L) = 3 > \max\{U_1(M, L), U_1(T, L)\} = 2.$$

$$U_1(B, C) = 8 > \max\{U_1(M, C), U_1(T, C)\} = 5.$$

$$U_1(B, R) = 5 > \max\{U_1(M, R), U_1(T, R)\} = 3.$$

3. Does the game have an equilibrium in dominant strategies (aka dominant strategy equilibrium)

Answer:

No. Player 2 has no dominant strategy; hence, there cannot be a dominant strategy equilibrium.

2.2 Nash Equilibrium

1. What is the Nash equilibrium in the 3x3 game above?

Answer:

Use the underlining method. In each *column*, underline the best outcome for player 1, and in each *row*, underline the best outcome for player 2. Is there a cell where both outcomes are underlined?

		Player 2		
		L	C	R
Player 1	T	1,2	2,1	3,4
	M	2,1	5,0	2,6
	B	3,7	8,2	5,5

*These notes are a consolidation of those of Jianmeng Lyu, Sam Leone, Pablo Munoz, Yassine Sbair Sassi, Jon Schellenberg, Nick Li, Katalin Springel, Jonas Tungodden, Dan Acland, Justin Gallagher, Mariana Carrera, Matt Leister, Ivan Balbuzanov, Anne Karing, and David Wu, Sam Wang, and Matteo Saccharola.

2. Find the pure strategy Nash equilibrium or equilibria in a game of Chicken.

Answer:

Use the underlining method. Is there a cell where both outcomes are underlined?

1\2	Veer	Don't Veer
Veer	2,2	1,3
Don't Veer	3,1	0,0

2.3 Mixed Strategy NE

1. Find the mixed strategy equilibrium for this Battle of the old GSIs.

Jon\Nick	BTS	Stray Kids
BTS	2,1	0,0
Stray Kids	0,0	1,2

Solution:

- Let's say Jon goes to BTS with probability j and Nick goes to BTS with probability n . We want to determine what values of j and n make $s^* = ((j, 1-j), (n, 1-n))$ a Nash equilibrium.
- First of all, what has to be true about the payoffs to Jon of choosing BTS versus Stray Kids if he is willing to randomize between the two?

Answer: He must be indifferent between playing either with probability 1, given the Nick's strategy $(n, 1-n)$.

- What has to be true about the payoffs of Nick choosing BTS versus Stray Kids if he is willing to randomize between the two?

Answer: He must be indifferent between playing either with probability 1, given Jon's strategy $(j, 1-j)$.

- Calculate payoffs for each pure strategy for Jon given that Nick is playing mixed strategy $s_n = (n, 1-n)$.

Answer:

Expected utility of BTS for Jon: $2n + 0(1-n) = 2n$.

Expected utility of Stray Kids for Jon: $0n + 1(1-n) = 1-n$

- Solve for the n that makes these expected payoffs equal.

Answer:

$$2n = 1 - n \Rightarrow n^* = 1/3$$

- Now to find the Jon's probability of going to Stray Kids, calculate the payoffs for Nick given that Jon is playing $s_j = (j, 1-j)$.

Answer:

Expected utility of BTS for Nick: $1j + 0(1-j) = j$

Expected utility of Stray Kids for Nick: $0j + 2(1-j) = 2-2j$

- Solve for the j that makes these expected payoffs equal.

Answer:

$$j = 2 - 2j \Rightarrow j^* = 2/3$$

- Now that we've found j so that Nick is willing to mix, and n so that Jon is willing to mix, we can write our mixed strategy Nash equilibrium.

Answer:

The mixed strategy Nash equilibrium is $(j^*, n^*) = (2/3, 1/3)$

- Notice that if either Nick or Jon deviates from this mixed strategy, the other will no longer be indifferent between the two pure strategies, and therefore, will choose the one that provides higher expected utility.